Utilization of Waste Mushroom-Sawdust Substrate as a Sustainable Alternative to Formaldehyde-Based Composites

Thomas Jefferson University
HOME OF SIDNEY KIMMEL MEDICAL COLLEGE

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Problem Statement and Background

- Widely used composite wood products – like particleboard and MDF – contain formaldehyde, which causes harm to the environment and human health [US Environmental Protection Agency, Center for Disease Control, American Cancer Society]
- Companies and countries are beginning to ban formaldehyde-based wood products, which presents an opportunity for bio-based materials
- Mycelium-based composites are a promising sustainable alternative [3] that incorporate agricultural waste materials
- Emerging heat-press techniques can result in rigid bio-composites with comparable mechanical performance [2]

Fabrication Approach –Heat Press Methodology

Heat-pressing parameters were refined to a temperature of 150°C for 15 minutes, after which samples were oven-dried to ensure complete desiccation.

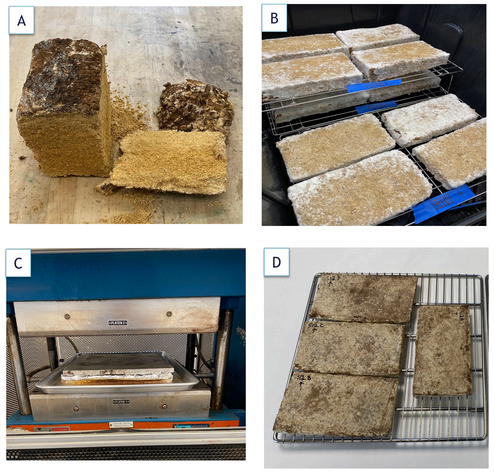
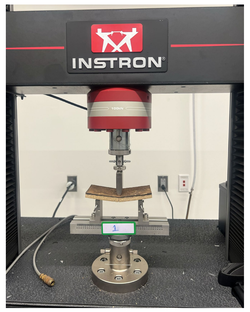
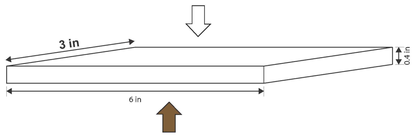


Figure 1: illustration of substrate preparation process. (A) Debarked and sliced substrates. (B) Regrown in conditioned environment. (C) Segmented and heat-pressed. (D) Post heat-press samples oven dried.

Material Performance Testing





Samples cut to specification were tested with load applied to both faces. A minimum of 12 specimens were tested for each face

Figure 2: Flexural strength testing was conducted following ASTM D1037-12 Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials on an Instron universal testing system.

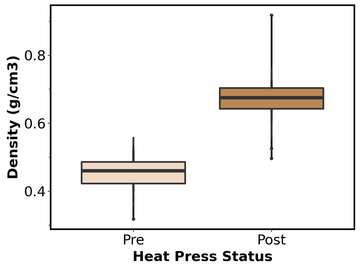
Opportunity: Waste Mushroom-Sawdust Substrate

Figure 3: Button mushroom growing on substrate, figure from [1]

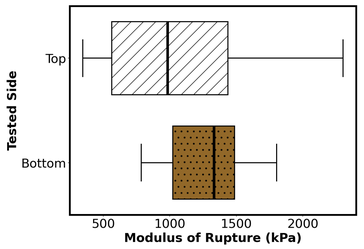
Edible mushrooms are grown on sawdust substrates. After the mushrooms are harvested, the remaining substrate (bottom portion) is currently a waste product

The waste material contains a mix of ligno-cellulosic plant fibers and mycelium, that together can form a strong bio-composite. Cellulose is an important structural component, and mycelium, the root-like structure of mushrooms, can act like a glue especially when heat-pressed.

Results



Material density pre- and post-heat-pressing reveal a clear difference with higher values for the post heat press samples. Statistical analysis demonstrates this difference is significant *($p < 0.05$)*. This disparity indicates that the heat pressing process increases the density of the material.



The graph displays a comparison of the modulus of rupture calculated from ASTM D1037 testing, comparing load applied from the top vs bottom of the samples. Both have overlapping ranges and close medians indicating a similar flexural behavior in each orientation. Statistically, there is no difference between the testing results suggesting that the material is isotropic.

SEM Imaging

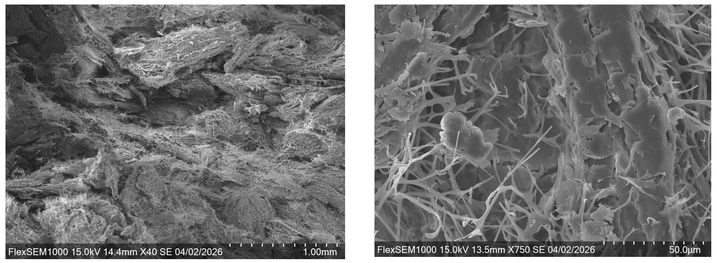


Figure 4: SEM images of mycelium samples cross-section parallel to the break. Sample (a): specimen at 40x magnification , Sample (b): specimen at 750x magnification

Objectives

This study utilizes waste substrates of shiitake (*Lentinula edodes*) sourced from a local Philadelphia mushroom supplier – Mycopolitan Mushroom Company to:

1. Develop a heat-press technique to make a reproducible bio-composite
2. Assess material performance including:
   - Bio-composite density changes across material states
   - Structural integrity, and the load bearing capacity in multiple directions (isotropic behavior)
3. Evaluate feasibility of production through considering scalability and potential to advance a regional circular economy

Key Findings and Next Steps

- Heat-pressing significantly increases density, creating a tightly-packed material
- SEM demonstrates there are few air pockets and void spaces in the highest density samples
- Mechanical testing shows that the material has isotropic behavior, with similar performance when tested on both faces of board

Figure 5: Mycelium Webbing (hyphae) of pre-heat-pressed sample. Taken on Keyence Digital Microscope

Figure 6: Comparison of post-heat-pressed sample next to existing product (Table)

- Next steps include reducing variability in material properties through investigating role of moisture and location within sample

Opportunities and Circular Economy

A biocomposite of mycelium and sawdust, formed through applied heat and pressure, has meaningful structural integrity. These novel materials create the possibility of a circular economy approach. Waste substrate can be sourced from local mushroom providers, diverting what would otherwise become costly waste to manage, and lead to material re-use, preventing the use of formaldehyde-based materials and their associated environmental harms.

Acknowledgements

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References

[1] Out-Grow LLC. (n.d.). White button mushroom (Agaricus bisporus). Retrieved April 8, 2026, from https://www.out-grow.com/products/white-button-mushroom-agaricus-bisporus

[2] Nussbaumer, M., Van Opdenbosch, D., Engelhardt, M., Briesen, H., Benz, J. P., & Karl, T. (2023). Material characterization of pressed and unpressed wood–mycelium composites derived from two Trametes species. Environmental Technology & Innovation, 30, 103063.

[3] Ahmad, W., McCormack, S. J., & Byrne, A. (2025a). Biocomposites for Sustainable Construction: A review of material properties, applications, research gaps, and contribution to circular economy. *Journal of Building Engineering*, *105*.